

SVETLOGRAD, Russian Federation

WMO: 349540

Lat: 45.350N

Lon: 42.850E

Elev: 141

StdP: 99.65

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-11.3	-18.4	0.8	-14.5	-14.7	1.1	-10.4	6.1	2.1	5.0	1.4	0.9	130	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	35.1	20.0	33.4	19.7	31.5	19.3	22.1	30.2	21.3	29.2	20.6	28.2	1.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.7	24.8	18.9	14.0	23.9	18.2	13.3	23.5	65.5	30.3	62.6	28.9	60.0	28.4	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.2	3.5	DB	-18.3	37.4	4.9	1.8	-21.8	38.6	-24.7	39.7	-27.5	40.7	-31.0	42.0	(4)
(5)			WB	-18.6	24.0	4.8	0.9	-22.1	24.6	-24.9	25.1	-27.6	25.6	-31.2	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-0.9	0.4	4.9	11.5	17.5	21.4	24.7	23.8	18.2	11.9	5.0	0.4	(6)	
	DBStd	10.02	4.87	5.26	4.24	4.28	3.89	3.20	2.75	2.86	3.78	4.89	4.41	5.17	(7)	
	HDD10.0	1302	339	270	166	32	2	0	0	0	1	37	156	299	(8)	
	HDD18.3	3009	597	503	417	206	64	10	0	1	48	206	399	557	(9)	
	CDD10.0	1892	0	1	7	77	233	341	455	429	247	95	6	0	(10)	
	CDD18.3	558	0	0	0	2	37	102	197	172	43	6	0	0	(11)	
	CDH23.3	5739	0	0	1	33	389	952	2168	1778	358	59	0	0	(12)	
	CDH26.7	2374	0	0	0	7	98	344	1039	790	84	12	0	0	(13)	
(14) Wind	WSAvg	1.3	1.3	1.5	1.7	1.4	1.2	1.2	1.0	1.0	1.1	1.0	1.3	1.4	(14)	
(15) Precipitation	PrecAvg	483	26	22	30	44	56	70	51	47	33	44	33	30	(15)	
	PrecMax	693	54	50	85	121	105	137	220	137	70	102	68	94	(16)	
	PrecMin	301	1	3	3	10	22	22	6	0	2	2	7	2	(17)	
	PrecStd	103	13	14	20	31	24	38	49	37	20	29	19	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	14.8	21.3	27.1	31.0	35.0	37.6	36.7	31.2	28.1	18.5	12.3	(19)	
		MCWB	6.5	8.3	11.5	15.3	17.9	19.5	20.3	20.5	19.1	16.5	12.3	8.5	(20)	
	2%	DB	7.7	11.4	16.8	23.5	29.1	32.5	35.9	34.6	29.0	24.7	14.8	9.8	(21)	
		MCWB	5.2	7.0	9.9	14.3	17.2	19.4	20.1	20.1	18.2	16.0	11.1	7.0	(22)	
	5%	DB	6.0	8.5	14.0	21.5	27.5	30.4	34.0	32.9	27.2	21.8	12.7	8.1	(23)	
		MCWB	4.0	5.2	8.4	13.3	17.0	18.9	19.6	19.7	17.7	15.2	10.0	6.0	(24)	
	10%	DB	4.5	6.3	11.6	19.3	25.5	28.5	32.1	31.0	25.3	18.9	10.9	6.7	(25)	
		MCWB	3.1	4.1	7.3	12.2	16.4	18.5	19.4	19.7	17.2	13.9	8.9	5.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	9.2	11.9	16.5	20.1	22.7	22.7	23.4	21.0	18.0	13.2	9.0	(27)	
		MCDB	9.7	13.9	19.5	24.2	27.1	30.5	32.0	30.7	27.5	24.8	16.2	12.0	(28)	
	2%	WB	5.5	7.1	10.4	15.3	19.0	21.2	21.7	22.3	19.7	16.8	11.7	7.4	(29)	
		MCDB	7.3	10.7	15.9	22.3	26.0	28.9	30.2	30.1	25.8	22.5	14.3	9.5	(30)	
	5%	WB	4.2	5.5	9.0	14.1	18.0	20.1	21.1	21.6	18.9	15.7	10.2	6.2	(31)	
		MCDB	5.7	7.9	12.9	20.0	24.8	27.4	29.8	29.2	24.8	20.6	12.2	7.9	(32)	
	10%	WB	3.2	4.2	7.8	12.8	17.1	19.3	20.4	20.8	18.1	14.6	9.0	5.2	(33)	
		MCDB	4.4	6.2	11.3	17.8	23.4	26.3	28.9	27.8	23.9	18.5	10.9	6.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.8	8.1	10.5	11.1	11.0	12.1	12.0	11.2	9.3	6.6	5.9	(35)	
		MCDBR	8.1	10.7	13.6	14.0	14.8	14.7	15.4	15.1	14.0	14.0	10.9	8.0	(36)	
	5% WB	MCWBR	5.9	6.6	7.5	6.3	5.6	5.1	4.2	4.5	5.4	6.5	6.8	5.6	(37)	
		MCWBR	7.2	9.9	11.7	12.4	12.2	12.6	12.1	11.5	12.3	9.7	7.3	7.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.341	0.375	0.415	0.461	0.428	0.434	0.458	0.459	0.429	0.390	0.355	0.345	(40)	
		taud	2.252	2.172	2.118	2.029	2.214	2.241	2.185	2.172	2.217	2.288	2.320	2.259	(41)	
	Ebn at Noon		739	782	803	798	843	838	811	793	782	760	721	689	(42)	
		Edh at Noon	85	112	135	161	138	135	141	138	122	99	80	77	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.14	2.00	2.96	4.63	5.80	6.45	6.43	5.79	4.22	2.52	1.34	0.92	(44)	
	RadStd		0.27	0.34	0.31	0.45	0.58	0.36	0.37	0.28	0.41	0.33	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A	

Nomenclature: See separate page